

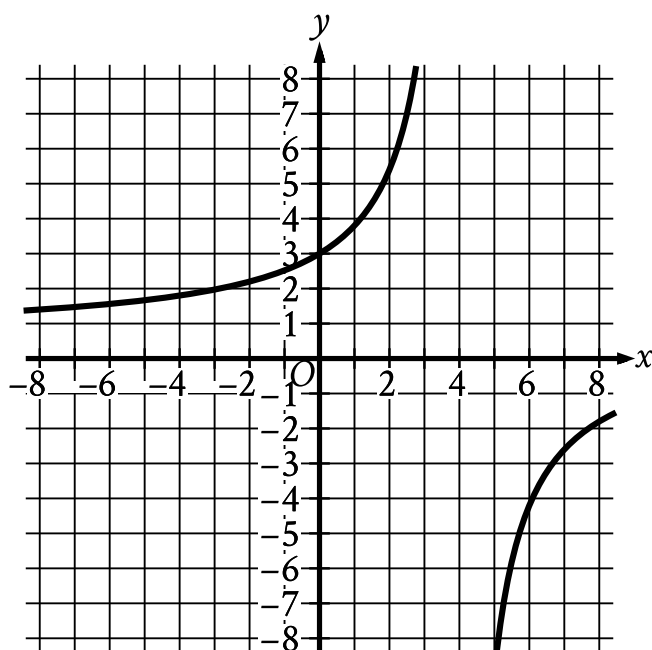
● EASY

● ADVANCED MATH

Nonlinear functions

10 questions · ~15 min

05



- For the first curve:
 - Moving from left to right:
 - The curve passes from quadrant 2 to quadrant 1.
 - In quadrant 2, the curve trends up gradually to point (0 comma 3).
 - In quadrant 1, the curve trends up sharply.
 - As x increases, the curve approaches the line x equals 4.
 - As x decreases, the curve approaches the line y equals 1.
 - The curve passes through the following points:
 - (negative 3 comma 2)
 - (0 comma 3)
- For the second curve:
 - Moving from left to right:
 - The curve is in quadrant 4.
 - In quadrant 4, the curve trends up sharply.
 - As x increases, the curve approaches the line y equals 1.

- As x decreases, the curve approaches the line x equals 4.
- The curve passes through the following point:
 - (6 comma negative 4)

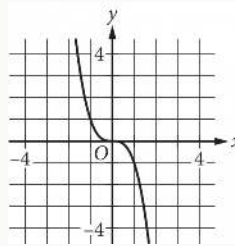
The graph of $y = f(x)$ is shown in the xy -plane. What is the value of $f(0)$?

- A. -3
- B. 0
- C. $\frac{3}{5}$
- D. 3

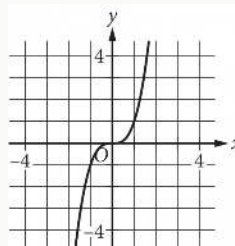
x	y
0	0
1	1
2	8
3	27

The table shown includes some values of x and their corresponding values of y . Which of the following graphs in the xy -plane could represent the relationship between x and y ?

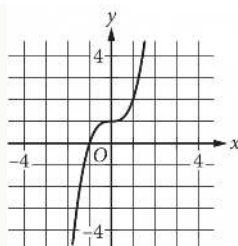
A.



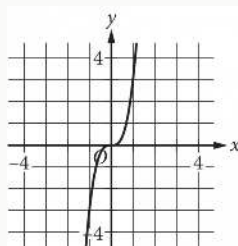
B.

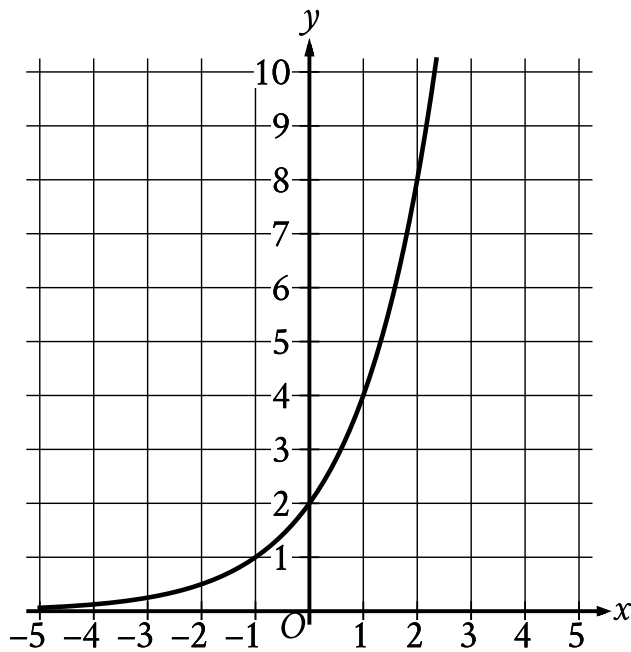


C.



D.





- Moving from left to right:
 - The curve passes from quadrant 2 to quadrant 1.
 - In quadrant 2, the curve trends up gradually to point (0 comma 2).
 - In quadrant 1, the curve trends up sharply.
- The curve passes through the following points:
 - (negative 1 comma 1)
 - (0 comma 2)
 - (2 comma 8)

What is the y-intercept of the graph shown?

- A. 0,0
- B. 0,2
- C. 2,0

D. 2, 2

Q4

ADVANCED MATH

The function f is defined by $fx = 10x^2 - 32x - 152$. What is the value of $f0$?

A. -152

B. -32

C. 0

D. 10

Q5

GRID-IN ADVANCED MATH

The function f is defined by $fx = x^2 + x + 71$. What is the value of $f2$?

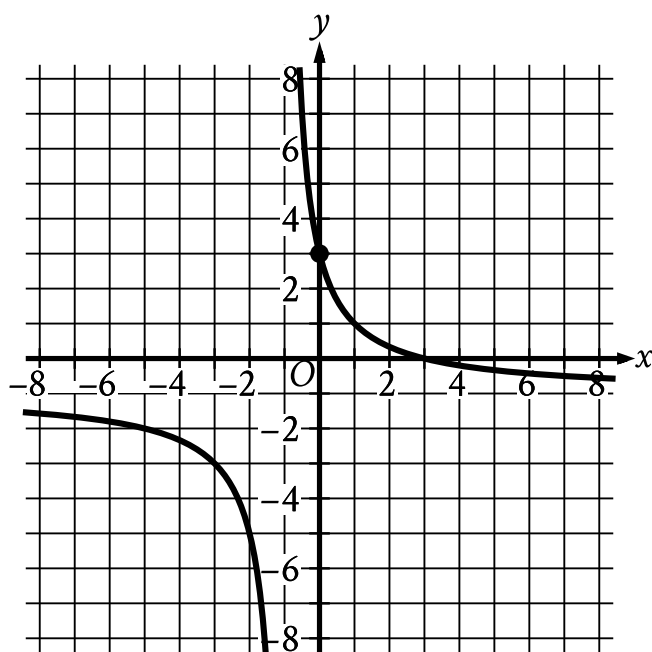
STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

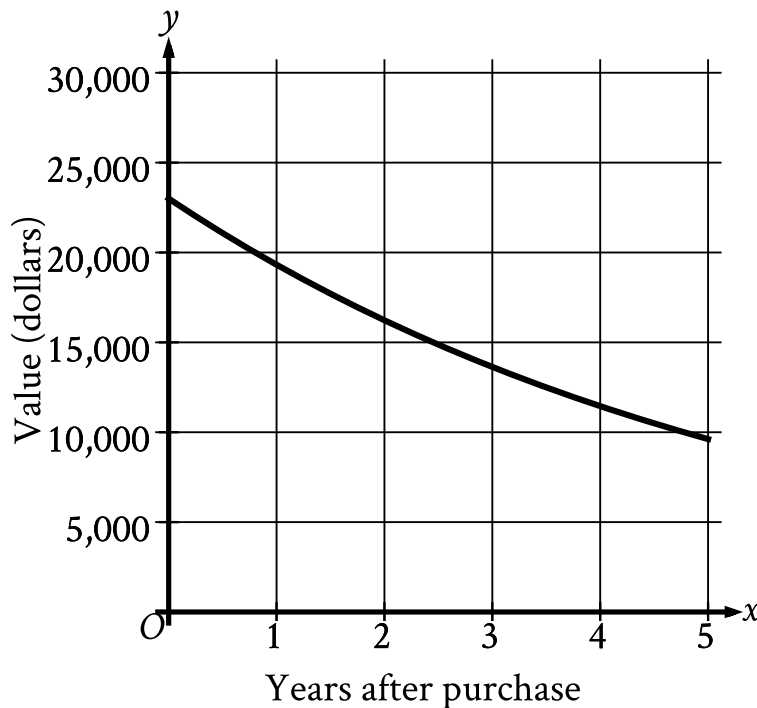
The function g is defined by $g(x) = |x + 18|$. What is the value of $g(4)$?

- A. -18
- B. -4
- C. 14
- D. 22



- For the first curve:
 - Moving from left to right, the curve is in quadrant 3.
 - As x decreases, the curve approaches the line y equals negative 1.
 - As x increases, the curve approaches the line x equals negative 1.
- For the second curve:
 - Moving from left to right, the curve passes from quadrant 2 to quadrant 1 to quadrant 4.
 - As x decreases, the curve approaches the line x equals negative 1.
 - As x increases, the curve approaches the line y equals negative 1.
- The 2 curves pass through the following points:
 - (negative 6 comma negative nine fifths)
 - (0 comma 3)

The graph shows the predicted value y , in dollars, of a certain sport utility vehicle x years after it is first purchased.



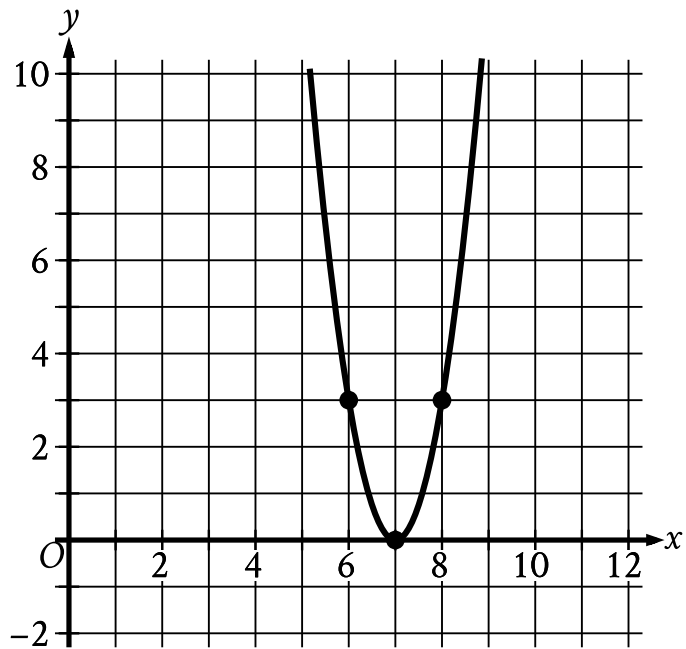
- The curve is in quadrant 1.
- The curve trends down gradually from left to right.
- The curve passes through the following approximate points:
 - (0 comma 23,000)
 - (1 comma 19,320)
 - (3 comma 13,632)
 - (5 comma 9,619)

Which of the following is closest to the predicted value of the sport utility vehicle 3 years after it is first purchased?

- A. \$ 9,619
- B. \$ 13,632

C. \$ 19,320

D. \$ 23,000



- The parabola opens upward.
- The vertex is at point (7 comma 0).
- The parabola passes through the following points:
 - (6 comma 3)
 - (7 comma 0)
 - (8 comma 3)

The x -intercept of the graph shown is $x, 0$. What is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
(0)	(0)	(0)	(0)
(1)	(1)	(1)	(1)
(2)	(2)	(2)	(2)
(3)	(3)	(3)	(3)
(4)	(4)	(4)	(4)

5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q10

ADVANCED MATH

At the time of posting a video, a social media channel had 53 subscribers. Each day for five days after the video was posted, the number of subscribers doubled from the number the previous day. Which equation gives the total number of subscribers, n , to the channel d days after the video was posted?

- A. $n = 53^d$
- B. $n = 53(2)^d$
- C. $n = 53\frac{1}{2}^d$
- D. $n = 53^2 + d$